MILITARY SPECIFICATION

BALLISTICS COMPUTER SUBASSEMBLY: 8635275 (OUTPUT UNIT)

1. SCOPE

1.1 This specification covers one type of electrical mechanical assembly known as Ballistics Computer Subassembly: 8635275, hereinafter referenced as "Output Unit" (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926 —Fire Control Materiel: General Specification Governing the Manufacture and Inspection of

MIL-P-14232 —Parts Equipment and Tools for Ordnance Materiel: Packaging of

MIL-I-45208 —Inspection System Requirements

MIL-I-45607 —Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 —Quality Assurance Terms and Definitions

MIL-STD-202 —Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

F8635275 —Ballistics Computer Subassembly: 8635275 (Output Unit)

IEL 8635275 —List of Inspection Documents for Ballistics Computer Subassembly: 8635275 (Output Unit)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication. The output unit shall be manufactured in accordance with Drawing F8635275 and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.3 General specification. The contractor shall be responsible for adherence to and compliance with the following requirements of MIL-F-13926:

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(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 **Environmental.**

3.4.1 *Operating temperatures.* The output unit shall meet all the requirements of 3.6 and 3.7 while exposed and thermally stabilized at ambient temperatures of −40° and +150° Fahrenheit (F). Upon return to standard ambient temperature the output unit shall meet the requirements of 3.4.2.

3.4.2 *Vibration.* The output unit shall meet the requirements of 3.5 to 3.7 inclusive after being vibrated in a vertical plane for five minutes plus or minus 15 seconds at an amplitude of 1/16 inch (⅛ inch total excursion) at a frequency of 30 cycles per second.

3.4.3 *Water resistance (sealing).* The interior of the output unit shall show no evidence of water penetration or other damage after being subjected to a simulated rainfall for a time duration of 15 minutes ± 15 seconds.

3.5 **Electrical.**

3.5.1 *Continuity.* The continuity of the output unit shall conform to the Schematic Drawing C10516305.

3.5.2 *Dielectric strength.* There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when the insulated areas of the output unit are subjected to an alternating current (ac) root mean square (rms) potential of 300 volts plus twice the rated voltage at a frequency of 60 cycles per second (cps) for a period of not more than 30 seconds nor less than five seconds.

3.5.3 *Insulation resistance.* All insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 10 megohms with 500 volts dc applied.

3.6 Performance.

3.6.1 *Output shaft rotation.* The output shaft for the ballistics drive shall rotate through an excursion of at least 12.94 rotations.

3.6.2 *Counter excursion.* The elevation correction counter shall cover an excursion from at least minus 1.0 mil to plus 63.7 mils when the requirement of 3.6.1 is met.

3.6.3 *Counter indication.* The counter shall indicate the angular position of the output shaft for the ballistics drive in a relationship that one revolution of the output shaft is equal to 5.0 mils as read on the counter within a tolerance of 0.05 mils.

3.6.4 *Superelevator output.* The superelevator output shaft shall make one revolution for each 6.25 mils of counter indication.

3.6.5 *Counter backlash.* Backlash between the output shaft for the ballistics drive and the counter shall not exceed 0.05 mils.

3.6.6 *Variable resistor backlash.* Backlash between the variable resistor and the output shaft for the ballistics drive shall not exceed 0.025 mils as measured at the output shaft.

3.6.7 *Output voltages resistor, R301.* The output voltage ratios of resistor R301 shall agree with the values listed in table I, within the tolerances specified for the ballistics drive output shaft positions of table I.

TABLE I. *Output Voltage Ratios*

Output shaft Position turns	Mils output (reference only)	Output ratio	Ratio tolerance
0	0	.000000	±.000900
2	10	.156528	±.000900
4	20	.312549	±.000900
6	30	.468515	±.000900
8	40	.624875	±.000900
10	50	.781374	±.000900
12	60	.937200	±.000900

3.6.8 *Output voltage noise, R301.* The maximum ac noise component is measured across a 250,000 load resistor shall not exceed .010 volts when the ballistics drive output shaft is turned from stop to stop in both di-

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rections at a rate of one mil per second with R301 excited with 26.0 volts direct current (dc).

3.6.9 *Motor.* Tachometer fixed phase excitation. Apply 115 volts plus or minus 1 volt 400 cycle ac between pins A and B of J301 for the fixed phase excitation of the motor-tachometer.

3.6.9.1 *Motor control phase excitation.* When energized as specified in 3.6.9, the application of 48 volts ± .5 volts 400 cycles ac between pins L and M of J301 shall cause the output shaft for the ballistics drive to turn counterclockwise (increasing output) when the voltage applied to pin L of J301 is in phase with the voltage applied to Pin A of J301, the output shaft for the ballistics drive shall turn clockwise (decreasing output) when the voltage applied to pin L of J301 is 180 degrees out of phase with the voltage applied to pin A of J301.

3.6.9.2 *Slew speed.* With the voltages applied as specified in 3.6.9 and 3.6.9.1 the output unit shall slew from stop to stop in both directions within a maximum running time of 6 seconds for each direction without load on output shaft and at standard temperature.

3.6.10 *Output torque.* With the voltages applied as specified in 3.6.9 and 3.6.9.1 the output unit shall be capable of driving under a 7 inch pound load applied to the ballistics drive output shaft.

3.6.11 *Tachometer output voltage.* Under the slew speed conditions of 3.6.9.2, the tachometer output voltage as measured at pin D of J301 with respect to pin K of J301 (ground) shall be a minimum of 4.0 volts ac, and in phase with the voltage applied to pin L of J301 within a tolerance of 30°.

3.7 **Illumination.** The mil counter when illuminated from a 24 volt power source shall be legible when observed in a darkened area.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or purchase order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *General provisions.* The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of MIL-I-45208 and the component inspection requirements of MIL-F-13926. Definitions of inspection terms not otherwise defined herein shall be as listed in MIL-STD-109.

4.2 **Inspection provisions.**

4.2.1 *Inspection lot.* Unless otherwise specified by the contracting officer, inspection lot size formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 *Examination and tests.* Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926 and the supplementary Quality Assurance Provisions listed on IEL 8635275. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with MIL-STD-105, and the inspection level and sampling plan specified in Table II titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Govern-

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ment reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

TABLE II. *Classification of Defects*

Use Level II of Table I with
Sampling Plan Table II–A of MIL–STD–105

	Requirement	Test procedure
CRITICAL: NONE		
MAJOR: AQL 1.0%		
101. Physical failure, post vibration	3.4.2	4.5.1.2
102. Continuity	3.5.1	4.5.2
103. Output shaft rotation	3.6.1	4.5.5
104. Counter excursion	3.6.2	4.5.5
105. Counter indication	3.6.3	4.5.5
106. Superelevator output	3.6.4	4.5.6
107. Counter backlash	3.6.5	4.5.7
108. Variable resistor backlash	3.6.6	4.5.8
109. Output voltage—R301	3.6.7	4.5.8
110. Output voltage—noise	3.6.8	4.5.9
111. Motor control—phase excitation	3.6.9.1	4.5.10.1
112. Slew speed	3.6.9.2	4.5.10.2
113. Output torque	3.6.10	4.5.11
114. Tachometer output voltage	3.6.11	4.5.12
115. Illumination	3.7	4.5.13
MINOR: NONE		

NOTE. The above tests shall be performed at standard ambient temperature (60° to 90° F.).

4.2.3 *Disposition of non-conforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL–STD–105.

4.3 **Control tests.**

4.3.1 *General.* One output unit shall be selected at random from each one hundred produced, or from each month's production, whichever occurs first. The control sample shall meet the following requirements and tests:

Control test	Requirement	Test procedure
301. Materials	3.1	As per spec, or dwg.
302 Fabrication	3.2	Visual—per dwg.
303. General specification	3.3	Visual—MIL–F–13926

4.3.2 *Environmental and electrical.* Three output units shall be selected at random from each fifty produced, or from each month's production, whichever occurs first. The control sample shall meet the following requirements and tests:

Control test	Requirement	Test procedure
304. Operating temperature	3.4.1	4.5.1.1
305. Water resistance	3.4.3	4.5.1.3
306. Dielectric strength	3.5.2	4.5.3
307. Insulation resistance	3.5.3	4.5.4

4.3.3 *Control test failure.* Should any one item of the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 **Inspection equipment.** Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Measuring and Test Equipment" requirement of MIL–I–45208 and all applicable requirements specified in MIL–I–45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 *Contractor design.* The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of MIL–F–13926. Concept, construction, materials, dimensions and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2 *Inspection equipment accuracy requirements.* The devices required in the test

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methods and procedures herein, to measure the electrical values specified in Section 3 shall provide readings to the accuracy specified below;

Item number	Description
1. Ratio Standard	Ratio range from .00000 to 1.00000 accuracy to ± .01%. Input impedance at 400 cps, 10,000 ohms minimum. Series impedance—10 ohms maximum. Phase shift 0.01° maximum.
2. Oscilloscope	Minimum sensitivity of 20 millivolts, full scale, vertical channel. External synchronization. Input impedance—1 megohm ± 10%.
3. A.C. Power Supply	0–130 volts rms, 400 cps ± 20 cps, 5% maximum harmonic distortion, 3 amperes output.
4. D.C. Power Supply	0–30 volts, maximum ripple—1.0 volt peak to peak, 300 milliampere output.

4.5 Test methods and procedures.

4.5.1 *Environmental.*

4.5.1.1 *Operating temperatures.* This test shall be conducted in conformance with the temperatures specified in 3.4.1. The test equipment shall be in accordance with the "Test Facilities" requirements specified in MIL–F–13926. While thermally stabilized at the specified temperatures the output unit shall be subjected to the tests specified in 4.5.5 to 4.5.13 inclusive to determine conformance with the requirements referenced therein.

4.5.1.2 *Vibration.* To determine conformance to 3.4.2 the output unit shall be mounted in its normal position using a suitable mounting apparatus to attach it to a vibrating machine or fixture, with provisions for securing universal assembly. Test equipment shall be in accordance with the "Test Facilities" portion of MIL–F–13926 and shall be capable of providing vibration at the amplitude and frequency for the time specified in 3.4.2. After vibration, the output unit shall be given a visual-tactile examination for damage and loose parts. The output unit shall then be subjected to the tests specified in 4.5.2 to 4.5.13 inclusive to determine conformance with the respective requirements referenced therein.

4.5.1.3 *Water resistance.* To determine conformance to 3.4.3, this test shall be conducted in accordance with Procedure I of MIL–F–13926 except for the time duration specified in 3.4.3. The output unit shall show no evidence of water penetration as a result of this test when visually examined.

4.5.2 *Continuity.* The output unit shall be tested, using standard measuring equipment, to demonstrate continuity of the circuits in accordance with Schematic Drawing D8635307 to determine compliance with 3.5.1.

4.5.3 *Dielectric strength.* To determine conformance to 3.5.2, the lamp shall be removed from the unit and the unit shall then be tested in accordance with Method 301 of MIL–STD–202.

4.5.4 *Insulation resistance.* To determine conformance to 3.5.3, with the lamp removed, the unit shall be tested to Method 302 of MIL–STD–202.

4.5.5 *Output shaft rotation and counter test.* To determine conformance to 3.6.1 through 3.6.3, the output shaft shall be slowly rotated from one stop to the other and the total excursion measured to establish a minimum of 12.94 turns. During this excursion, the elevation counter shall be observed by a means of magnification for a relationship of 5 mils counter-scale reading to one revolution of the output shaft. While the output shaft is traversing through its total excursion of 12.94 turns, the mil-counter shall read from minus 1.0 mil to plus 63.7 mils.

4.5.6 *Superelevator output test.* To determine conformance to 3.6.4, the superelevation shaft shall be slowly rotated from one stop to the other while the elevation counter is observed as in 4.5.5 for a relationship of 6.25 mils counterscale reading to one revolution of the superelevation shaft.

4.5.7 *Counter backlash test.* To determine conformance to 3.6.5 the counter scale shall

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be observed for no movement through a telescope containing a reticle line while the angular rotation of the output shaft is checked for .05 mil maximum backlash.

4.5.8 *Output voltage of R301 and variable resistor backlash tests.* To determine conformance to 3.6.6 and 3.6.7 attach an ac ratio standard (4.4.2) in parallel with resistor R301 at pins E and H of connector J301. Place an oscilloscope (4.4.2) into the test set up by connecting to pin G of connector J301 and the ratio standard slider. Connect a 250,000 ohm resistor between pins G and E of connector J301. Apply 25.6 volts ac 400 cps between pins E and H of connector J301 and observe the oscilloscope for a near null reading when the ratio standard slider is moved to obtain the settings listed in table I and the output shaft is at the corresponding position. An adjustment of the ratio slider, within the tolerance limits specified in table I will be permitted to bring the scope to a null reading. Output units which null with out-of-tolerance ratio settings shall be rejected. During each of the readings, determine the blacklash of the variable resistor with relation to the output shaft. Approach the null point from one direction, observe the output shaft position, proceed through the null point then return to null point. The angular backlash of the output shaft shall not exceed .025 mils.

4.5.9 *Output voltage noise test.* To determine conformance to 3.6.8, connect a 250,000 ohm resistor between pins F and G of connector J301 and impress a 26 volt dc potential across pins F and H. Turn the ballistics drive output shaft from stop to stop in both directions at a rate of one mil per second while measuring the ac noise component across the 250,000 resistor. The noise component shall not exceed .010 volt.

4.5.10 *Motor—tachometer fixed phase excitation.* To determine conformance to 3.6.9, apply 115 volts ac 400 cps between pins A and B of connector J301 and observe the fixed phase excitation of the motor-tachometer. This condition shall be utilized for the following tests.

4.5.10.1 *Motor control phase excitation test.* To determine conformance to 3.6.9.1, apply 48 volts ac 400 cps between pins L and M of connector J301 and 115 volts ac 400 cps to pins A and B. Observe the phase relationship of the voltages at pins A and L on an oscilloscope, and observe the direction of rotation of the output shaft. When L voltage is in-phase with A the output shaft shall turn counterclockwise. When L voltage is 180° out of phase with pin A the output shaft shall turn clockwise.

4.5.10.2 *Slew speed test.* To determine conformance to 3.6.9.2 apply 115 volts and 48 volts at the pins specified in 4.5.10 and 4.5.10.1. With the voltage at pin L in phase with Pin A and at a maximum, observe and time the slewing action of the output shaft as it moves from stop to stop. With no load on the output shaft, and at standard temperature, the shaft shall complete the full excursion in six seconds. Repeat the test in the opposite direction with the voltage at pin L 180° out-of-phase with pin A.

4.5.11 *Output shaft torque test.* To determine conformance to 3.6.10, attach a 7 inch pound torque load to the output shaft and apply the same voltages specified in 4.5.10.2. The output shaft shall slew from stop to stop under this load. Repeat the test in the opposite direction.

4.5.12 *Tachometer output voltage test.* To determine conformance to 3.6.11, while conducting the slew speed test of 4.6.10.2 measure the voltage at pin D of connector J301 with respect to pin K of J301. This voltage shall be a minimum of 4.0 volts ac and should be approximately in-phase with the voltage applied at pin L.

4.5.13 *Illumination test.* To determine conformance to 3.7, apply 24 volts dc to pins K and C of connector J301. With the output unit in a darkened area the counter shall be clearly visible.

5. PREPARATION FOR DELIVERY

5.1 **Packaging, packing and marking.** Packaging, packing and marking shall be in ac-

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cordance with MIL–P–14232 and the applicable packaging data sheet. The level of protection shall be as specified in the procurement document (see 6.2).

6. NOTES

6.1 Intended use. The output unit is primarily intended as a functional component of the electronic analog Computer, Ballistics: M16. The output unit is designed to convert electrical correction signals from the ballistics computer into shaft rotation for use in the ballistics drive and gun superelevator of tank fire control systems.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of an applicable level of preservation, packaging and packing. (See 5.1)
(c) Applicable Packaging Data Sheet Number. (See 5.1)
(d) Applicable stock number.

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